MATH PAPER 1

Mock Examination — Paper 1

Paper 1 (Long Questions)

135 minutes (2.25 hours)

Total marks：105

FOR REVISION

For revision and self-assessment only. This is not an official examination paper.

Difficulty rating：**3.5** / 5.0

INSTRUCTIONS

This paper consists of Sections A and B. Answer ALL questions and show all working. This is an AI-generated unofficial mock exercise.

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＊Independent, unofficial mock paper aligned to the DSE syllabus. Not affiliated with any official examination authority. For revision only.

甲部**(1)**

1. (a) Factorize $x^2 + 2x - 15$.

   (b) Hence, factorize $x^2 + 2x - 15 - 4xy - 20y$.

(3 marks)

2. (a) Simplify $\left(\frac{8x^{-3}}{27y^6}\right)^{-\frac{1}{3}}$, expressing the answer with positive indices only, where x and y are positive numbers.

   (b) Hence, find the value of the expression when $x = 4$ and $y = \frac{1}{3}$.

(3 marks)

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3. (a) The sum of an arithmetic series is $S = \frac{n}{2}[2a + (n-1)d]$, where a is the first term, d is the common difference and n is the number of terms with n greater than 1. Make d the subject of the formula.

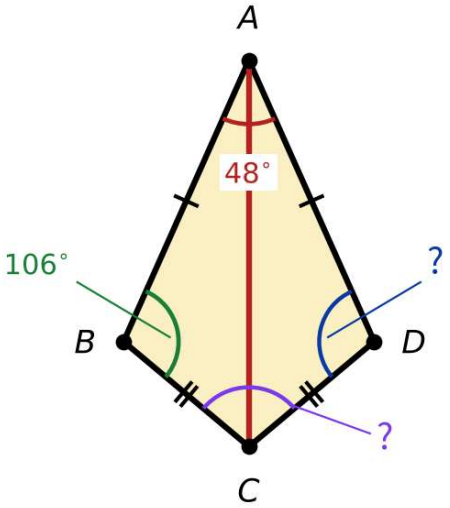
(b) Hence, find the value of d when $S = 185$, $a = 5$ and $n = 10$.

(3 marks)

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4.



(a) In the figure, $ABCD$ is a kite-shaped kite frame in which $AB = AD$ and $CB = CD$. It is given that $\angle BAD = 48°$ and $\angle ABC = 106°$. Find $\angle ADC$ and $\angle BCD$.

(b) Hence, find $\angle ACD$ and $\angle CAD$.

(4 marks)

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5. (a) A person deposits a sum of money in a bank account at an interest rate of **5%** per annum, compounded yearly. The interest obtained after **2** years is **3075** dollars. Find the sum of money.

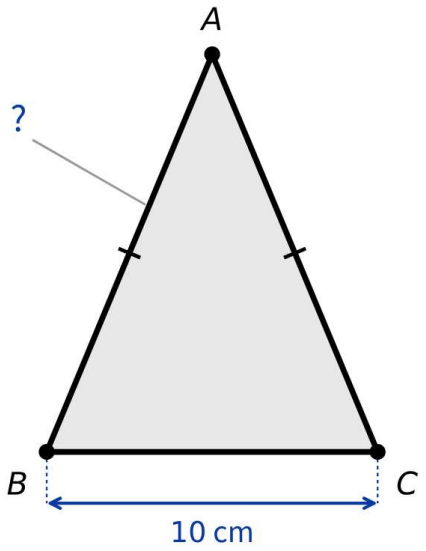
(b) Hence, find the amount of the sum of money after it has been deposited in the same account for **3** years, correct to the nearest dollar.

(4 marks)

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6.



(a) In the figure, $\triangle ABC$ is an isosceles triangular metal plate in which $AB = AC$. It is given that the perimeter of the plate is **36** cm and $BC = 10$ cm. Find AB.

(b) Find the area of $\triangle ABC$.

(4 marks)

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7. (a) The median of the five data $\mathbf{3}$, $\boldsymbol{x}$, $\mathbf{8}$, $\mathbf{12}$ and $\mathbf{15}$ is $\mathbf{9}$, where $\boldsymbol{x}$ is an integer. Find the value of $\boldsymbol{x}$.

(b) Hence, find the mean and the range of the five data.

(4 marks)

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8. (a) The length of a rectangular aluminium board is **7** cm longer than its width and the area of the board is **120** cm^2. Find the width of the board.

(b) Hence, find the length of a diagonal of the board.

(5 marks)

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9. (a) In the rectangular coordinate plane, the equation of the straight line L_1 is $2x + y - 11 = 0$ and the equation of the straight line L_2 is $x - y - 1 = 0$. L_1 and L_2 intersect at the point P . Find the coordinates of P .

(b) Hence, find the equation of the straight line which passes through P and is perpendicular to L_1 , giving the answer in general form.

(5 marks)

甲部(2)

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10. (a) The pressure P of the gas in a sealed container varies directly as its temperature T and inversely as its volume V. When $T = 300$ and $V = 5$, $P = 120$. Express P in terms of T and V.

(b) Find the value of P when $T = 450$ and $V = 6$.

(c) If the temperature of the gas is increased by 20% and its volume is decreased by 25%, find the percentage increase in its pressure.

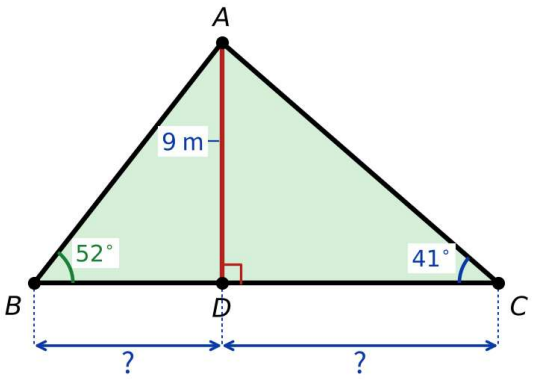
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11.



(a) In the figure, $\triangle ABC$ is a triangular garden and AD is the perpendicular from A to BC, where D lies on BC. It is given that $AD = 9$ m, $\angle ABC = 52°$ and $\angle ACB = 41°$. Find the lengths of BD and CD, correct to **3** significant figures.

(b) Hence, find the length of BC and the area of the garden, correct to **3** significant figures.

(c) Find the length of AB and the distance between the point D and AC, correct to **3** significant figures.

(7 marks)

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12. (a) The sum of the first n terms of an arithmetic series is $S_n = 3n^2 + 5n$, where n is any positive integer. Find the first term and the 2nd term of the series.

(b) Hence, find the common difference and the nth term of the series.

(c) Find the value of n such that S_n exceeds 1000 for the first time, and the nth term of the series.

(7 marks)

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13. (a) Seismologists describe the scale of an earthquake by its magnitude on the Richter scale. As the energies released by different earthquakes may differ enormously, comparing the values of the energy directly is inconvenient, so seismologists define the magnitude using the common logarithm of the energy. Let E joules be the energy released by an earthquake and M be its magnitude on the Richter scale, where $\log E = 4.8 + 1.5M$ and M is a positive number. The magnitude of an earthquake on the Richter scale is 6 . Find the energy released by this earthquake, giving the answer in scientific notation correct to 3 significant figures.

(b) The energy released by another earthquake is 2×10^{15} joules. Find the magnitude of this earthquake on the Richter scale, correct to 1 decimal place.

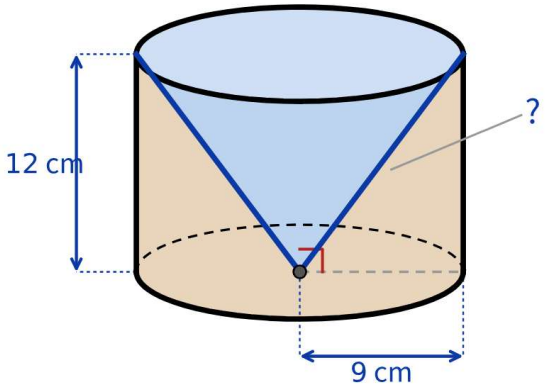
(c) Someone claims that if the magnitude of an earthquake on the Richter scale is 2 higher than that of another earthquake, then the energy released by the former is 1000 times that released by the latter. Do you agree? Explain your answer, stating in your explanation whether this multiple depends on the original magnitude.

(7 marks)

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14.



(a) The ornament in the figure is made by removing a cone from the top of a solid cylindrical block of wood. The base radius of the cylinder is **9** cm and its height is **12** cm. The base of the cone removed coincides with the top face of the cylinder and the vertex of the cone is at the centre of the base of the cylinder. Find the volume of the cylinder in terms of π.

(b) Find the volume of the cone removed and hence the volume of the ornament, giving the answers in terms of π.

(c) Find the slant height of the cone removed and hence the curved surface area of that cone, giving the latter in terms of π.

(d) Find the total surface area of the ornament, correct to **3** significant figures.

(8 marks)

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乙部

15. (a) It is given that a and b are real numbers and $(a+bi)(3-2i)=13+13i$, where i is the imaginary unit and $i^2=-1$. Find a and b.

(b) Hence, find the real part and the imaginary part of $(a+bi)^2$.

(5 marks)

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16. (a) It is given that the quadratic equation $x^2 + (k-3)x + k = 0$ has no real roots, where k is a real number. Find the range of values of k.

(b) Hence, find the greatest integral value and the least integral value of k for which the equation has no real roots.

(5 marks)

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17. (a) Simplify $\sin^4\theta - \cos^4\theta$, expressing the answer in terms of $\sin\theta$ only.

(b) It is given that $\sin\theta = \frac{3}{5}$. Hence, find the value of $\sin^4\theta - \cos^4\theta$.

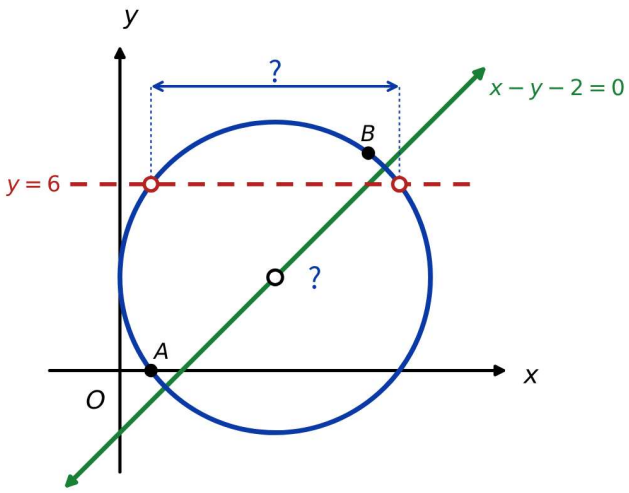
(c) Solve the equation $\sin^4\theta - \cos^4\theta = -\frac{1}{2}$, where $0° \leq \theta \leq 360°$.

(6 marks)

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18.



(a) In the figure, the circle C passes through $A(1, 0)$ and $B(8, 7)$, and the centre of C lies on the straight line $x - y - 2 = 0$. Find the coordinates of the centre and the radius of C.

(b) Hence, write down the equation of C, giving the answer in general form.

(c) Find the length of the chord cut on the circle C by the straight line $y = 6$.

(7 marks)

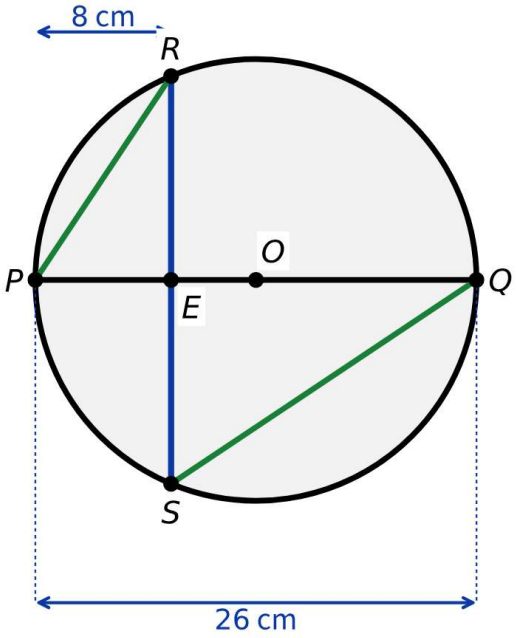
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19.



(a) A workshop is installing two straight steel cables on a circular metal plate for reinforcement. In the figure, PQ and RS are two chords of the plate representing the two cables, and they intersect at a point E inside the plate. It is given that PQ is a diameter of the plate and the centre of the plate is O. The engineer of the workshop points out that once the two triangles formed at E are shown to be similar, the remaining segment can be found from the lengths of the other three. Prove that $\triangle PER \sim \triangle SEQ$.

(b) It is given that the diameter PQ of the plate is 26 cm and the distance between P and the intersection E is 8 cm. All the following sub-parts refer to this plate and these two cables, and all lengths are in cm.

(12 marks)

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